SAMPLE

# Great Observatories: The Compton Gamma Ray Observatory

The Gamma Ray Sky

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## Introduction

Of the four great space observatories the American space agency built to cover the sky's full spectrum, the one this volume treats was the wildest: a seventeen-ton machine built to catch the most violent light there is, gamma radiation, the kind that does not pass through a window or a window's worth of air, the kind that arrives from exploding stars, devouring black holes, and the mysterious flashes that were, at the time this machine was designed, the biggest unsolved puzzle in high-energy astronomy. It rode to orbit on a space shuttle because no rocket of its day could lift it, it worked for nine years on a beat far above the atmosphere that blocks its quarry, and when its pointing machinery wore out, it was steered down by command into a remote ocean corridor, the largest human object ever brought home on purpose. It was named for the physicist whose discovery first proved that light itself can behave as a particle, and the naming was exact, because that discovery is the working principle of every detector the machine carried.

The volume follows the catalog's standing method for observatories: the machine as the protagonist, its instruments as its organs, its findings as its deeds, and its operating years as a working career with a beginning, a middle, an honorable end, and a successor. The reader who has met the earlier volumes in this series knows the questions that structure the chapters: what was built, why the builders chose those answers, what the machine actually measured and how, which findings changed the textbooks, what went wrong and what stayed stubborn, what came after, and the small delights of the story that no fact sheet captures.

The preview paragraph for readers in a hurry: launched in the spring of 1991 as the second of the four great observatories; four instruments spanning the gamma band from the soft edge at tens of kilo-electron-volts to giga-electron-volt energies a million times higher; the burst watcher that recorded over two thousand of the sky's mystery flashes and proved they come from far beyond the galaxy; the telescope that discovered whole classes of objects shining in the hardest light, including the discovery that certain quasars pour out gamma rays and that the sky's gamma map is crowded; the spectrometer that mapped the galaxy's radioactive aluminum and the glow of annihilating positrons near the galactic center; and a planned, commanded descent into the Pacific at the mission's end, the first deliberate re-entry of so large a spacecraft, so that the machine's final act was itself an engineering first. The chapters unpack each clause.

## Chapter 1: The Discovery of The Compton Gamma Ray Observatory

The observatory's discovery story runs in three movements, and the first begins long before the spacecraft, with the physics of the man it is named for. In the early 1920s an American experimenter, scattering X-rays off electrons in matter, found that the scattered light came out with less energy than it went in, exactly as if the light were a stream of particles colliding with the electrons like billiard balls. The result, which earned him the physics prize of 1927, settled a decades-old argument about the nature of radiation in favor of the particle picture, and it gave the new field of gamma-ray detection its working physics: to catch a gamma ray, let it scatter off an electron and measure the electron. The observatory that carried the physicist's name carried his scattering in every detector it flew, which is a rare exactness in a monument's naming.

The second movement is the sky's. Gamma rays are stopped by the atmosphere entirely, which is excellent for the reader, who would not otherwise be reading, and terrible for the astronomer, who cannot build an observatory for them on the ground. The first hints of cosmic gamma rays came from the very edge of space: balloon and satellite instruments in the 1960s and early 1970s, some of them in truth nuclear-test-monitoring satellites repurposed for science, recorded flashes of gamma radiation with no obvious local source, and a set of findings that became the young field's founding puzzle: the flashes were coming from everywhere, lasting seconds or less, and nothing on the sky's map matched them. The theorists' proposals ranged from comet collisions in the solar neighborhood to wars between distant civilizations, which the field records fairly as a measure of the genuine confusion. What the field needed was a machine with enough collecting power to count the flashes properly, watch them steadily, and map where they do and do not come from.

The third movement is institutional, and it is the part that made the machine what it was. The American space agency, having decided in the 1970s to build a suite of four great orbiting observatories covering the spectrum from visible light through ultraviolet, X-ray, and gamma ray, assigned the gamma band its own flagship and, in the commissioning of the fleet, named it after the scattering physicist. The requirements that shaped the design were written directly from the young field's puzzles: to catch the mystery flashes in honest statistics, the burst instrument should look at the whole sky at once; to identify the steady sources, the telescopes should map and measure

spectra across the widest possible span of energies; and to do all of it from orbit, the machine should carry its four instruments together, share one pointing, and live long enough for the sky's slow statistics to accumulate.

The machine's own name carries a small piece of the discovery story. The fleet's first sibling carried the name of the astronomer who discovered the galaxies' escape; the gamma sibling carried the name of the physicist whose scattering the detectors ran on; and the naming convention, which the modern reader can meet in any press release of the era, told the public what each machine was for before a word of its science was published. The catalog records the habit approvingly: a named machine is a legible machine, and this one's name doubled as its physics lesson, which the science chapter cashes in full.

The flight up was the discovery story's public climax. The observatory was too heavy for the era's conventional rockets, so it rode in the shuttle's cargo bay, seventeen tons of instruments and supporting structure, and it went up in the spring of 1991, deployed from the orbiter's arm by a crew that included, the trivia chapter records, an astronomer who would become the fleet's most famous public face. The deployment day held one genuine scare, a stuck antenna boom that an unplanned spacewalk freed by hand, a reminder that the era's spacecraft were flown by people as well as by procedures. Released into its working orbit, a low path around the Earth at an angle that swept every latitude of the sky, the machine checked out its four instruments, opened its observing campaign, and the discovery phase ended: the observatory was on station, and the gamma sky, seen properly for the first time, began to arrive.

The two small pioneers deserve the discovery story's own note, because the flagship's requirements were drafted from their frustrations. Before the big machine, the gamma sky was watched by two modest satellites: the American small-astronomy mission of the early 1970s, which lived months and found the first gamma pulsars and the galaxy's plane glowing in hard light, and its European successor later in the decade, which flew a spark chamber like the flagship's own and catalogued a couple of dozen sources in the machine's own band. Both missions ran on counts of hundreds and thousands of photons, the statistics the flagship would later turn into hundreds of thousands, and the field's requirements documents quoted their limits as the design's motivation: more collecting area, more years, more of the sky watched at once. The catalog's note is the inheritance's honesty: the great observatory was not a leap from ignorance but a scaling-up of demonstrated instruments, the pioneers having proven the detectors and merely run out of lifetime before the sky's statistics were due.

The machine's own biography carries a naming footnote the reader should collect now. Launched as the Gamma Ray Observatory, a plain bureaucratic title, it flew for a year under the acronym before the space agency renamed it in orbit for the scattering physicist, the christening made official in a ceremony the mission's veterans still recount; the renaming was the fleet's habit, the siblings having picked up their astronomers' names the same way, and the honor was posthumous in this case, the physicist having died decades before his machine flew. The catalog's footnote within the footnote: the renaming set a small precedent, that a spacecraft's name is a decision about how the public will remember it, and the agency's choice of a pure physicist rather than an astronomer told the truth about the mission: it was less a telescope than a particle-physics laboratory flown for astronomy, the naming being the program's one act of self-description, accurate to the experiment.